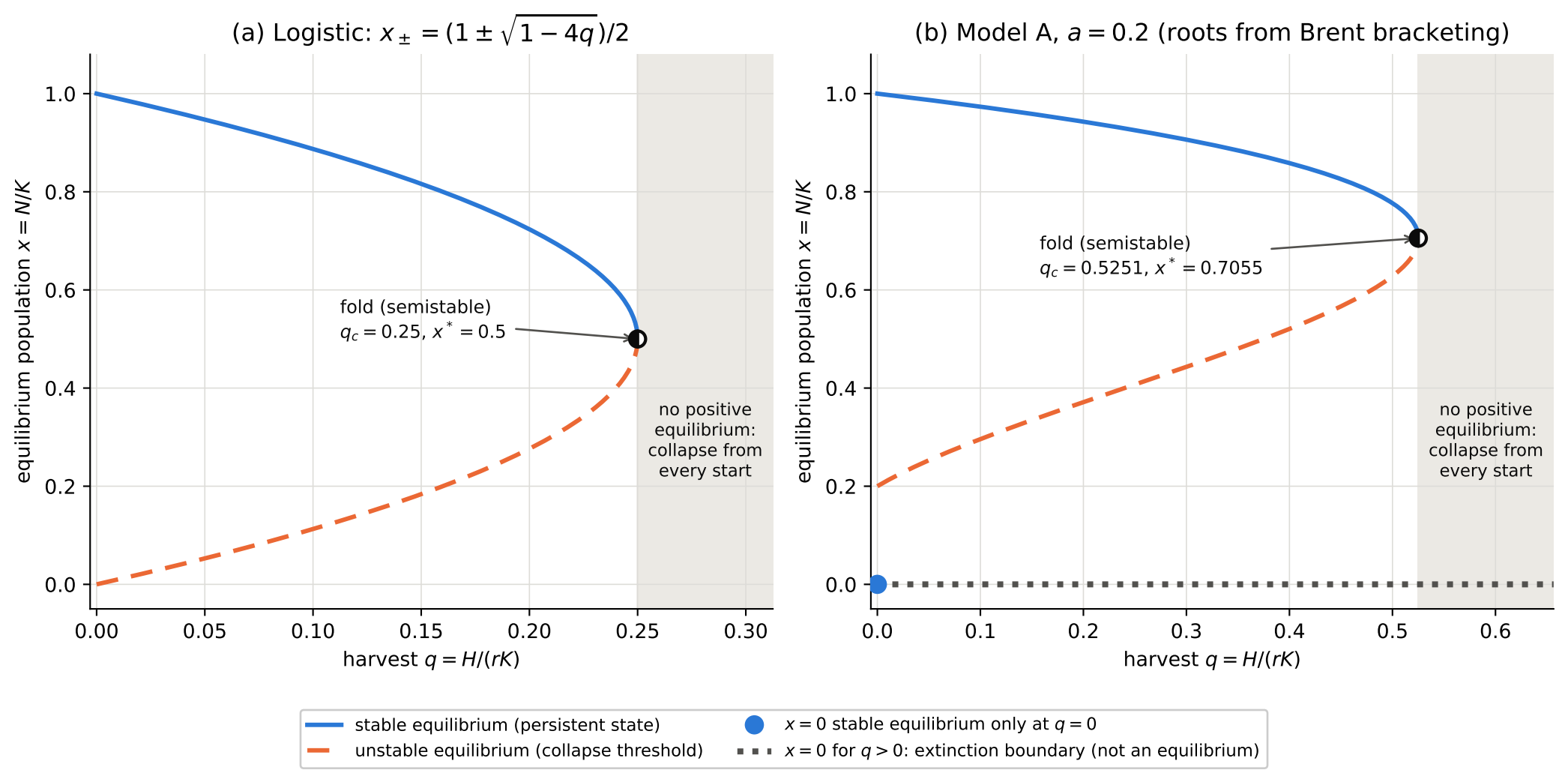
(a) Logistic: $x_{\pm} = (1 \pm \sqrt{1-4q})/2$
(b) Model A, $a = 0.2$ (roots from Brent bracketing)
equilibrium population $x = N/K$
harvest $q = H/(rK)$
fold (semistable)
$q_c = 0.25$, $x^* = 0.5$
fold (semistable)
$q_c = 0.5251$, $x^* = 0.7055$
no positive equilibrium: collapse from every start
no positive equilibrium: collapse from every start
stable equilibrium (persistent state)
unstable equilibrium (collapse threshold)
$x = 0$ stable equilibrium only at $q = 0$
$x = 0$ for $q > 0$: extinction boundary (not an equilibrium)